

during stock selection. A brief history of these approaches is described below.

First Wave: The first wave of portfolio optimization with transaction costs focused on incorporating the bid-ask spread cost into the optimization process. The belief was that, since trading costs are generally lower for large cap liquid stocks and trading costs are generally higher for small cap illiquid stocks, spreads would be a good proxy for costs since spreads are generally lower for large cap stocks and higher for small cap stocks. By decreasing expected returns by the round-trip spread cost, managers felt that the optimizer would determine a more appropriate mix of stock and a more accurate expected return. The optimized solution would apply larger weights to stocks with smaller spreads and lower weights to stocks with higher spreads. While this process was a good first step in the process it still did not account for the possibility that a large number of shares of a large cap liquid stock could, in fact, be more expensive than a small number of shares of a small cap illiquid stock. The first wave of portfolio optimization with transaction costs did not account for the cost associated with the size of the order.

Second Wave: The second wave of portfolio optimization with transaction costs focused on incorporating a market impact estimate that was dependent upon order size. These types of models have been previously formulated by Balduzzi and Lynch (1999) and by Lobo, Fazel, and Boyd (2006). In this process, larger orders will have higher market impact cost than smaller orders in the same names. The expectation is that the optimization process would determine sizes that could be easily absorbed into the market without incurring inappropriate levels of impact and the resulting optimal solution would provide a more efficient allocation of dollars across the different stocks. In this approach, however, the market impact formula used was based on a “static” cost-size relationship and does not provide any cost reduction benefits from the underlying execution strategy.

This means that estimated cost will be exactly the same for the number of shares transacted regardless of whether those shares were to be transacted with a high level of urgency or passively through the day. Furthermore, the optimization process, even though it considered the risk term to determine the optimal mix of stocks and portfolio weightings, does not consider the risk composition of the other names in the trade list to determine corresponding impact cost. For example, suppose the trade list consists of only a single buy order for 500,000 shares of RLK. The manager is exposed to both market risk and company specific